

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457202.6365 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.067 +/- 0.045
Transit depth (Rp/Rs)^2	0.45 +/- 0.61 [%]
Orbital Inclination (inc)	83.29 +/- 2.88
Airmass coefficient 1 (a1)	0.725883 +/- 5.6e-05
Airmass coefficient 2 (a2)	0.17 +/- 0.22
Scatter in the residuals of the lightcurve fit is	0.0028 %
Best Comparison Star	#3 - [91, 452]
Optimal Aperture	49.22
Optimal Annulus	143.23
Transit Duration (day)	0.011 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.067 ± 0.045 vs 0.1326 (deviation 1.46σ)
Duration fit vs published	0.011 d vs 0.0737 d (deviation 5.23σ)
Mid-time O–C	-3090.7 min
Depth SNR	1.5
Residual scatter	0.0028 %

Light curve

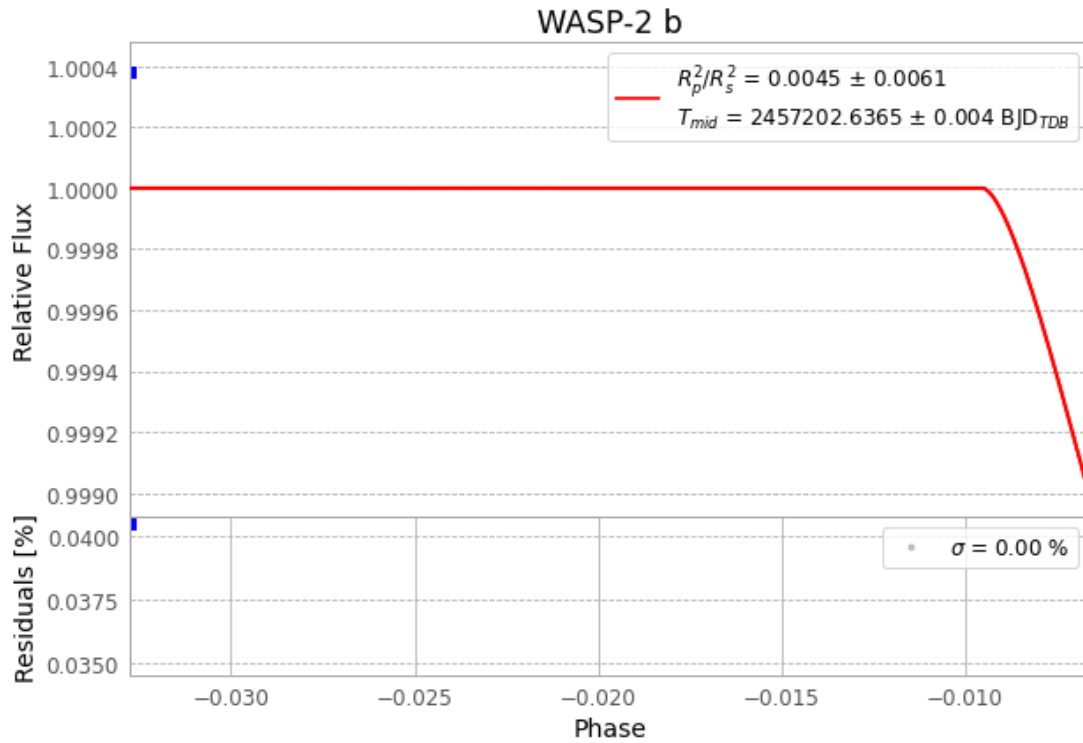


Figure 1 — FinalLightCurve_WASP-2 b_2015-06-30.png (SHA-256 90c9ad040b372195...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [387, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b68955907c03c81d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-2150701045112.FITS → WASP-2150701083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-150701.log (dabe4ff6aed6...), FinalLightCurve_WASP-2 b_2015-06-30.png (90c9ad040b37...), FinalLightCurve_WASP-2 b_2015-06-30.csv (bb7658051917...)

Compute resources

Wall time 1172.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:07Z.